



CSE471: System Analysis and Design
Assignment on Functional Requirements
Proposed Project Title: A Personalized Decision Support
Platform for Study Abroad and Scholarship Planning

Group No: 2, CSE471 Lab Section: 7 , Summer 2026		
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Project Overview

A web-based study abroad and scholarship decision support platform that helps students make informed decisions about international higher education. The platform provides personalized university and scholarship recommendations based on a student's academic profile, budget, career goals, and preferences. It automatically collects and updates information from official university websites, scholarship portals, and other trusted public sources to ensure accurate and up-to-date data. In addition to university matching and scholarship eligibility, the system offers features such as application tracking, deadline monitoring, funding analysis, visa preparation, and personalized guidance through an AI-assisted advisor and public insight engine. The platform is managed by **Students**, **Administrators**, and **Content Managers**, creating a centralized and reliable ecosystem for study abroad planning.

Tech Stack:

- Language: Typescript , Javascript
- Framework: [React.js](https://reactjs.org/), Express.js
- Styling: TailwindCSS
- Database: PostgreSQL
- ORM: Prisma
- **Authentication:** JWT / Google OAuth
- **Search:** PostgreSQL Full-Text Search
- **Vector Search:** pgvector
- **Deployment:** Vercel / Render / Railway

- External APIs:

1. Gemini API / OpenAI API

For:

- AI study abroad advisor
- Public insight summarization
- Recommendation explanation

2. Gmail API / Resend API

For:

- Deadline reminders
- Scholarship alerts
- Requirement change notifications
- Verification emails

3. Open Maps

4. Exchange rate data api

User Roles

1. Student

The primary user of the system. Students can create academic profiles, search for universities, receive personalized university and scholarship recommendations, compare countries, track applications, manage documents, view deadlines, access public insights, and use the AI study abroad advisor.

2. Content Manager

Responsible for managing and verifying educational data. Content Managers can review crawled university data, scholarship information, visa resources, public insight sources, deadline updates, and requirement changes before publishing them to users.

3. Admin

The system super-user. Admins can manage users, approve or remove content, monitor system activity, manage university and scholarship databases, control crawler activity, review reports, manage subscriptions if added later, and maintain overall platform quality.

Functional Requirements

Module 1

Member	Feature Title	Description
1	Candidate Readiness Scorecard	Instead of just storing data, the system instantly analyzes the user's profile to generate a "Readiness Score" for specific tiers of universities. It highlights the user's competitive advantages (e.g., research papers) and flags potential weaknesses (e.g., low IELTS score) to show where they need improvement before applying.
2	Smart University Matching	The system uses the "Readiness Score" from Feature 1 to automatically categorize university programs into Safe, Target, or Reach buckets based on the user's specific academic history, ensuring recommendations are tailored to their actual probability of acceptance.
3	Scholarship Eligibility Engine	This engine focuses strictly on financial matching. It cross-references the student's profile (nationality, research interest, GPA) against exclusive scholarship databases to calculate a "Matching Percentage" for aid, ignoring irrelevant opportunities.
4	Country Decision Dashboard	This feature focuses on macro-environmental factors. It compares countries based on qualitative data (post-study work visas, cost of

		living, part-time job laws) rather than academic matching, helping the user decide <i>where</i> to apply based on life goals, not just grades.
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Module 2

Member	Feature Title	Description
1	Application Strategy Builder	After university matching, the system suggests a balanced application plan, such as 3 Safe, 4 Target, and 2 Reach universities. This helps students reduce risk and avoid applying only to highly competitive or unsuitable universities.

2	Smart Document Checklist	For each selected university or country, the system generates a personalized checklist of required documents such as passport, transcript, IELTS certificate, SOP, LOR, CV, bank statement, and visa documents. Students can mark documents as pending, prepared, or submitted.
3	Application Progress Tracker	Students can track every university application from planning to enrollment. Statuses may include Planned, Preparing, Applied, Interview, Offer Received, Rejected, Visa Processing, and Enrolled.
4	Deadline & Requirement Monitor	The system monitors application deadlines, scholarship deadlines, and admission requirement changes. If a deadline is

		near or a requirement changes, such as IELTS 6.0 becoming 6.5, the student receives an alert.
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Module 3

Member	Feature Title	Description
1	Public Insight Engine	The system collects information from trusted public sources such as university FAQs, embassy pages, public blogs, public forums, and YouTube transcripts. It summarizes common advice, warnings, student experiences, and country-specific insights with source references.
2	University Reality Check	This feature shows practical information that official university pages often do not

		highlight, such as housing difficulty, hidden costs, part-time job availability, language barriers, student satisfaction, and accommodation challenges.
3	Similar Student Finder	Students can compare their profile with anonymized previous applicant profiles. For example, a student with CGPA 3.2 and IELTS 6.5 can see where similar students applied, whether they were accepted, and whether they received scholarships.
4	AI Study Abroad Advisor	The AI advisor answers personalized questions using the student profile and platform database. It can explain why a university is suitable, compare countries, summarize public insights, and suggest next steps, but it does not replace the rule-based matching system.

Module 4

Member	Feature Title	Description
1	Cost of Degree Calculator	The system calculates the estimated total cost of studying abroad, including tuition, living expenses, visa fee, insurance, application fee, flight cost, and emergency funds. Students can adjust values to create realistic budget plans.
2	Funding Gap Analyzer	This feature compares the student's available budget with the estimated total cost. If there is a funding gap, the system suggests scholarships, lower-cost universities, cheaper countries, or alternative application options.
3	Visa Preparation Hub	Students get country-specific visa guidance, including

		required documents, financial proof, embassy links, processing timeline, visa fees, and common mistakes. This helps students prepare after receiving an offer letter.
4	Personalized Opportunity Feed	The dashboard shows personalized updates based on the student profile, such as new scholarships, upcoming deadlines, matching universities, visa updates, requirement changes, and public insights relevant to their target country or subject.